

The answer key for the **General** round is below.

p	Answer	p	Answer
1	18%	11	13
2	8	12	65
3	2	13	3780
4	51	14	1233
5	3	15	18
6	134π	16	363
7	120	17	2
8	$\frac{10}{7}$	18	45
9	29	19	72
10	465	20	6

Problem 1. The probability that Angie has an umbrella given that it is raining is 30%. The probability that it rains is 60%. What is the probability that Angie has an umbrella, and it is raining? Express your answer as a percentage.

Proposed by: Sneha Kundu

Answer: 18%

Solution: We can use the following formula where Event A is Angie has an umbrella and Event B is that it's raining.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

We are given that $P(A | B) = 30\%$ and $P(B) = 60\%$. We are looking for $P(A \cap B) = P(A | B) \cdot P(B) = 30\% \cdot 60\% = 18\%$.

Problem 2. Find the number of positive integers k such that $\frac{30}{k}$ is an integer.

Proposed by: Zack Berchenko

Answer: 8

Solution: Observe that the question is just asking for the number of positive divisors of 30. We know that $30 = 2^1 \cdot 3^1 \cdot 5^1$ so the number of divisors is $(1+1)(1+1)(1+1) = 2^3 = \boxed{8}$.

Problem 3. Rectangle $ABCD$ has side length $AB = 8$. Let E be the intersection of AC and BD . Given that $AE = 5$, find $\frac{[BCD]}{[AED]}$. Note that $[BCD]$ and $[AED]$ represent the areas of triangles BCD and AED , respectively.

Proposed by: Samuel Kretzschmar

Answer: 2

Solution: Noting that $[ABE], [BCE], [CDE], [AED]$ are all equal,

$$\frac{[BCD]}{[AED]} = \frac{[BCE] + [CDE]}{[AED]} = \boxed{2}.$$

Problem 4. Sneha, Evan, and Taison are comparing their ages. Sneha says, "My age seven years ago was exactly half of Taison's age seven years ago." Taison replies, "My age right now is the average of Sneha and Evan's ages." Finally, Evan says "I'm ten years older than Sneha." What is the sum of their ages?

Proposed by: Zack Berchenko

Answer: 51

Solution: Let a, b , and e denote the ages of Aaron, Brian, and Ethan, respectively. We then have equations

$$a - 7 = \frac{e - 7}{2} \tag{1}$$

$$e = \frac{a + b}{2} \tag{2}$$

$$b = a + 10 \tag{3}$$

Substituting equation (3) into equation (2) gives,

$$e = \frac{a + (a + 10)}{2} = a + 5.$$

Substituting this into equation (1) gives,

$$\begin{aligned} a - 7 &= \frac{(a + 5) - 7}{2} \\ a - 7 &= \frac{a}{2} - 1 \\ \frac{a}{2} - 7 &= -1 \\ a - 14 &= -2 \\ a &= 12. \end{aligned}$$

Thus, $e = 12 + 5 = 17$ and $b = 12 + 10 = 22$. Hence, the sum of their ages is $12 + 17 + 22 = 51$.

Problem 5. Find the smallest prime factor of $4^5 - 7^2$.

Proposed by: Zack Berchenko

Answer: 3

Solution: We see $4^5 - 7^2 = (2^5)^2 - 7^2 = (2^5 - 7)(2^5 + 7) = 25 \cdot 39 = 3 \cdot 5^2 \cdot 13$. Thus the smallest prime factor is 3.

Problem 6. Let Ω be a circle with center O , and let $\overline{BC}$ be a diameter of Ω . Let A be a point on Ω such that $\angle ABC = 45^\circ$. If the area of $\triangle AOC$ is 67, determine the area of Ω .

Proposed by: Zack Berchenko

Answer: 134π

Solution: $\angle AOC = 90^\circ$, so area of $\triangle AOC$ is $\frac{r^2}{2}$. Thus, since $\frac{r^2}{2} = 67$, the area of Ω is $\pi r^2 = (2 \cdot 67)\pi = 134\pi$.

Problem 7. Sam eats lunch at a cafeteria. He has three options for a main dish, four options for a side dish, and five options for dessert. If Sam wants to get one main dish, one side dish, and two desserts, how many distinct lunches could Sam eat? (It is not allowed to get the same dessert twice, and the order of desserts does not matter.)

Proposed by: Zack Berchenko

Answer: 120

Solution: We have three options for the main, four options for the side, and $\binom{5}{2} = 10$ options for dessert. Thus, the answer is

$$3 \cdot 4 \cdot 10 = 120.$$

Problem 8. Austin and Jeff run to an ice cream store at their own constant speeds. The ice cream store is a mile away from where Austin and Jeff start. Jeff gets there first, and spends fifteen minutes ordering and eating his ice cream. He then gets bored of waiting, and begins running back towards Austin. He meets Austin halfway between the ice cream store and their starting location. If Jeff runs at 15 mph, what is Austin's speed in mph?

Proposed by: Zack Berchenko

Answer: $\boxed{\frac{10}{7}}$

Solution: Jeff runs 1 mile to the ice cream store and 0.5 miles back, so in total, he runs 1.5 miles. At a rate of 15 mph, this will take him $\frac{1.5}{15}$ hours = 6 minutes. Since Jeff spends 15 minutes in the ice cream store, he is gone for 21 total minutes before meeting Austin. This means Austin runs 0.5 miles in 21 minutes, or $\frac{7}{20}$ of an hour. Hence, we can compute Austin's running speed, in mph, to be $\frac{0.5}{7/20} = \boxed{\frac{10}{7}}$.

Problem 9. Let a and b be positive numbers. Given that $a^2b^2 = 169$ and $3a + 3b = 27$, compute $(a - b)^2$.

Proposed by: Taison Scofield

Answer: $\boxed{29}$

Solution: We can take the square root of both sides of the first equation and divide the second equation by 3, getting $ab = 13$ and $a + b = 9$.

Next, square both sides of $a + b = 9$ to get $a^2 + 2ab + b^2 = 81$. Note that

$$\begin{aligned}(a - b)^2 &= a^2 - 2ab + b^2 \\ &= a^2 + 2ab + b^2 - 4ab \\ &= (a + b)^2 - 4ab \\ &= 81 - 4(13) \\ &= \boxed{29}.\end{aligned}$$

Problem 10. Let $ABCD$ be a trapezoid with $AB \parallel CD$. Given that $AB = 17$, $BC = 25$, $CD = 45$, and $DA = 17$, compute the area of $ABCD$.

Proposed by: Taison Scofield

Answer: $\boxed{465}$

Solution: We will present two solutions.

- Notice that this trapezoid can be divided into a $15 - 20 - 25$ right triangle, an $8 - 15 - 17$ right triangle, and a 17×15 rectangle. We get the area to be $150 + 60 + 255 = \boxed{465}$.
 - Remove the 17×15 rectangle in the middle, leaving a triangle with side lengths of 17, 25, and 28. Then use Heron's Formula to find the area of the triangle and add it to the area of the rectangle.
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Problem 11. Austin, Eric, Sam, Sneha, and Ryan are waiting in line to use the communal library computer to search for the books they want. Austin's search takes 4 minutes to complete, Eric's takes 8 minutes, Sam's takes 2 minutes, Sneha's takes 6 minutes, and Ryan's takes 3 minutes. If the five people are randomly arranged in line to the computer, and only one person can use it at a time, what is the expected number of minutes that will pass until Ryan completes his search?

Proposed by: Ryan Niu

Answer: 13

Solution:

Note for any ordering of the 5 people, there is one corresponding "reversed" ordering in which the combined wait time of the ordering and its reverse is $4+8+2+6+3+3$, since an ordering and its reverse will have exactly one waiting instance of each person, and 2 waiting instances of Ryan because he must wait for his 3 minute search regardless of the ordering. Thus, the average of this time will give us the expected value to be $\frac{4+8+2+6+3+3}{2} = \frac{26}{2} = \boxed{13}$.

Problem 12. The positive integer n satisfies the condition that $n^2 + n$ has 5 prime divisors and 32 positive divisors. Find the smallest possible value of n .

Proposed by: Jefferson Zhou

Answer: 65

Solution: Note that the condition is equivalent to $n^2 + n = n(n+1)$ being the product of 5 distinct prime factors. We now split into cases.

Case 1: Either n or $n+1$ is the product of 3 distinct primes, and the other one is the product of 2 distinct primes.

We can list the smallest numbers which are the product of 3 distinct primes: the ones which are divisible by 6 are 30, 42, 66, 78, and so on, where 66 yields the solution $n = 65$. The ones which are divisible by 10 but not 6 are 70, 110, and so on, the first of which yields the solution $n = 69$. There are no smaller solutions when the number which is a product of 3 distinct primes is divisible by 2. If this number is not divisible by 2, then the smallest solution is $n = 105$, which is too big. Thus, the smallest solution in this case is $n = 65$.

Case 2: Either n or $n+1$ is the product of 4 distinct primes, and the other one is a prime.

The smallest number which is the product of 4 distinct primes is 210, which indeed gives the solution $n = 210$. However, this number is too large.

There are no more cases, so the smallest value of n is 65.

Problem 13. Bob wants to color each of the numbers in the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ either red, green, or blue, so that there are at least three elements of each color, including one element which is a multiple of three. How many such colorings are possible?

Proposed by: Taison Scofield & Samuel Kretzschmar

Answer: 3780

Solution: First, note that there are only 3 multiples of three (3, 6, 9), so they must be different colors. There are therefore $3! = 6$ ways to color these three. Of the seven remaining, we see that since each color must have at least 3 elements, only one

$(10 - 3 \cdot 3 = 1)$ element is left after the required colorings. We choose one color to have this last element, which there are 3 choices for; then there are 4 elements of this color, and 3 elements of the other two. As the multiples of three gave each color one element already, we choose three elements for this color out of the remaining seven, which have $\binom{7}{3} = 35$ ways of being chosen, then two elements for another color out of the $7 - 3 = 4$ remaining, which have $\binom{4}{2} = 6$ choices. The final color automatically gets the remaining 2 elements. The number of ways to do this is therefore

$$6 \cdot 3 \cdot 35 \cdot 6 = \boxed{3780}.$$

Problem 14. For positive integers a , b , c , and d , we have

$$\frac{a}{2} - \frac{b}{3} + \frac{c}{5} - \frac{d}{7} = \frac{1}{210}.$$

Find the smallest possible value of $1000a + 100b + 10c + d$.

Proposed by: Samuel Kretzschmar

Answer: $\boxed{1233}$

Solution: Multiplying by 210, we get $3 \cdot 5 \cdot 7a - 2 \cdot 5 \cdot 7b + 2 \cdot 3 \cdot 7c - 2 \cdot 3 \cdot 5a = 1$. Taking these modulo 2, 3, 5, and 7, we get $105a \equiv a \equiv 1 \pmod{2}$, $-70b \equiv 2b \equiv 1 \pmod{3}$, $42c \equiv 2c \equiv 1 \pmod{5}$, and $-30d \equiv 5d \equiv 1 \pmod{7}$ respectively. This yields $a \equiv 1 \pmod{2}$, $b \equiv 2 \pmod{3}$, $c \equiv 3 \pmod{5}$, $d \equiv 3 \pmod{7}$; checking that indeed $\frac{1}{2} - \frac{2}{3} + \frac{3}{5} - \frac{3}{7} = \frac{1}{210}$, our answer is 1233.

Problem 15. Let r_1 and r_2 denote the roots of the polynomial $x^2 - 7x + 1$. Compute $r_1\sqrt{r_1} + r_2\sqrt{r_2}$.

Proposed by: Ryan Niu & Samuel Kretzschmar

Answer: $\boxed{18}$

Solution:

By Vieta's, we know that $r_1 + r_2 = 7$ and $r_1r_2 = 1$. We find

$$(r_1\sqrt{r_1} + r_2\sqrt{r_2})^2 = (r_1\sqrt{r_1})^2 + (r_2\sqrt{r_2})^2 + 2(r_1\sqrt{r_1})(r_2\sqrt{r_2}) = r_1^3 + r_2^3 + 2(1\sqrt{1}).$$

We see by sum of cubes that

$$r_1^3 + r_2^3 = (r_1 + r_2)(r_1^2 - r_1r_2 + r_2^2) = 7((r_1 + r_2)^2 - 3r_1r_2) = 7(7^2 - 3) = 322,$$

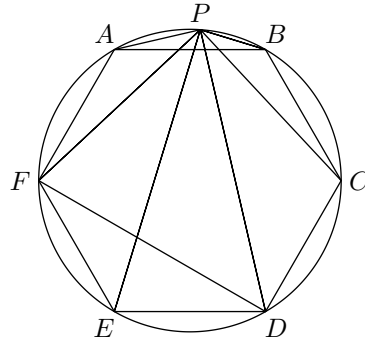
$$\text{so } r_1\sqrt{r_1} + r_2\sqrt{r_2} = \sqrt{r_1^3 + r_2^3 + 2} = \sqrt{324} = \boxed{18}.$$

Problem 16. Let $ABCDEF$ be a regular hexagon of side length 12 and let ω be its circumcircle. Point P lies on minor arc AB of ω such that $PE = 13\sqrt{3}$. Compute $PF \cdot PD$.

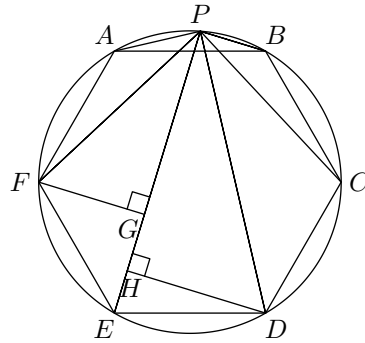
Proposed by: Jefferson Zhou

Answer: $\boxed{363}$

Solution:



Note that $\angle FPE = \angle FDE = 30^\circ = \angle EFD = \angle EPD$. Let G and H be the feet of the altitudes from F and D to PE , respectively.



Let $PF = x$ and $PD = y$. Then, $FG = \frac{x}{2}$, $PG = \frac{x\sqrt{3}}{2}$, $HD = \frac{y}{2}$, and $PH = \frac{y\sqrt{3}}{2}$. As such, $144 = FE^2 = FG^2 + GE^2 = \frac{x^2}{4} + \left(13\sqrt{3} - \frac{x\sqrt{3}}{2}\right)^2 = \frac{x^2}{4} + \frac{3x^2}{4} - 39x + 507$,

so $x^2 - 39x + 363 = 0$. Also, $144 = DE^2 = DH^2 + HE^2 = \frac{y^2}{4} + \left(13\sqrt{3} - \frac{y\sqrt{3}}{2}\right)^2 = \frac{y^2}{4} + \frac{3y^2}{4} - 39y + 507$, so $y^2 - 39y + 363 = 0$. Since the only two points on ω which are equidistant from D and F are B and E , we have $x \neq y$, so x and y are the roots of $a^2 - 39a + 363$, so by Vieta's, $xy = \boxed{363}$.

Note: This solution is basically just the Law of Cosines applied to triangles $\triangle FPE$ and $\triangle DPE$.

Alternate Solution: We will solve this generally. Let the side length of the hexagon be R . We use Ptolemy's on cyclic quadrilateral $PDEF$ to see

$$PD \cdot EF + PF \cdot ED = R \cdot PF + R \cdot PD = R\sqrt{3} \cdot PE = DF \cdot PE$$

so dividing by R and squaring both sides yields $PF^2 + 2PF \cdot PD + PD^2 = 3PE^2$ (1). As $\angle FPD = 60^\circ$ by the Inscribed Angle Theorem, we use Law of Cosines to get

$$PF^2 + PD^2 - 2 \cdot PF \cdot PD \cdot \cos(60^\circ) = PF^2 + PD^2 - PF \cdot PD = DF^2 = 3R^2. \quad (2)$$

Subtracting (2) from (1) gives $3PF \cdot PD = 3(PE^2 - R^2)$ so $PF \cdot PD = (PE^2 - R^2) = (13\sqrt{3})^2 - 12^2 = 363$.

Problem 17. Let f be a function such that $f(1) = 2$ and $f(n) = f(n-1)^2$ for all $n \geq 2$. Find the remainder when $f(f(f(3)))$ is divided by $2^7 - 1 = 127$.

Proposed by: Ryan Niu

Answer: $\boxed{2}$

Solution:

From the starting definition, we can deduce that $f(n) = 2^{2^{(n-1)}}$. Thus, $f(f(f(3))) = f(f(16)) = f(2^{2^{15}}) = 2^{2^{(2^{2^{15}}-1)}}$. Observe that $2^{2^{15}} - 1 \equiv (-1)^{2^{15}} - 1 \pmod{3} \equiv 0 \pmod{3}$. Thus, $2^{(2^{2^{15}}-1)} \equiv 1 \pmod{7}$ since $2^3 \equiv 1 \pmod{7}$. Finally, since $2^7 \equiv 1 \pmod{127}$, we get that $2^{(2^{2^{15}}-1)} \equiv 2^1 \cdot 1 \pmod{127} \equiv 2 \pmod{127}$. Thus, our final remainder is $\boxed{2}$. Another idea is to note that $f(n)$ has a period of 3 modulo 127, and is equal to 1 modulo 3 for $n \geq 2$, so $f(f(3)) \equiv 1 \pmod{3}$ and $f(f(f(3))) \equiv 2 \pmod{127}$.

Problem 18. Let S be the value of the infinite sum

$$\sum_{n=2025}^{\infty} \frac{n^2 - n - 1}{n!} = \frac{2025^2 - 2025 - 1}{2025!} + \frac{2026^2 - 2026 - 1}{2026!} + \cdots.$$

The number $N = 2025! \cdot S$ is an integer. Compute the number of positive divisors of N .

Proposed by: Samuel Kretzschmar

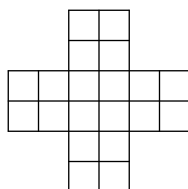
Answer: $\boxed{45}$

Solution: We find

$$\begin{aligned} \sum_{n=2025}^{\infty} \frac{n^2 - n - 1}{n!} &= \sum_{n=2025}^{\infty} \left(\frac{n(n-1)}{n!} - \frac{1}{n!} \right) \\ &= \sum_{n=2023}^{\infty} \frac{1}{n!} - \sum_{n=2025}^{\infty} \frac{1}{n!} \\ &= \frac{1}{2023!} + \frac{1}{2024!}. \end{aligned}$$

Thus $N = \frac{2025!}{2023!} + \frac{2025!}{2024!} = 2024 \cdot 2025 + 2025 = 2025^2 = 3^8 \cdot 5^4$, which has $(8+1)(4+1) = 45$ factors.

Problem 19. Consider the board below, consisting of a 6×6 square with the corner 2×2 squares removed. How many ways are there to completely tile the board using only 1×2 and 2×1 pieces?

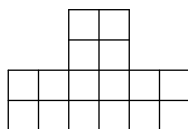


Proposed by: Angie Huang

Answer: $\boxed{72}$

Solution: Let S_n be the number of ways to tile a $2 \times n$ board with 2×1 and 1×2 pieces. Note that it is easy to see that $S_n = F_{n+1}$. Let a be the number of ways to tile our desired board, and let b be the number of ways to tile the board shown below. First, consider the bottom 2×2 square in the desired board. It can either be tiled

completely, in two ways, or with a 1×2 on the bottom and two 2×1 pieces filling in the remaining portion, in which case the remainder of the board has $S_2^2 \cdot S_3$ ways to tile. Thus, $a = 2b + 2 \cdot 2 \cdot 3 = 2b + 12$. Now, consider the upper 2×2 square in the board attached. It can either be tiled completely, in two ways, or with a 1×2 at the top and two 2×1 pieces filling in the remaining portion, in which case the remainder of the board has S_2^2 ways to tile. Thus, $b = 2S_6 + 2 \cdot 2 = 26 + 4 = 30$, and we obtain $a = 2 \cdot 30 + 12 = 72$.



Problem 20. Let $ABCDE$ be a pentagon inscribed in a circle with $AB = BC$. Let P be the intersection of chords AD and BE , Q be the intersection of chords BD and CE , and R be the intersection of chords BD and AC . Given that $AP = 3$, $PD = 9$, and $AR = 8$, determine the length of $\overline{PQ}$.

Proposed by: Sneha Kundu

Answer: 6

Solution: We note that since $AB = BC$, line BD bisects $\angle ADC$. We can angle chase to find $\angle PDQ = \angle ADB = \angle BDC = \angle BEC = \angle PEQ$. Since D and E are on the same side of PQ (opposite B), we find $PQDE$ is cyclic. Thus $\angle ACE = \angle ADE = \angle PDE = \angle PQE$; since Q is on CE , this implies that $AC \parallel PQ$. Therefore $\triangle DAR \sim \triangle DPQ$, so $\frac{PQ}{AR} = \frac{PD}{AD}$ and $PQ = 8 \cdot \frac{9}{3+9} = \boxed{6}$.

